

Dify Local RAG Setup Guide

A Practical Learning Journal

Disclaimer

This repository does not include any **Large Language Models (LLMs), Embedding Models, datasets, or copyrighted materials**.

If your dataset contains copyrighted materials (for example, company training manuals, internal documentation, university lecture slides, textbooks, or any other proprietary content), it is strongly recommended that you keep your RAG system local and do not publish or redistribute those materials.

The purpose of this project is to document the complete setup process, troubleshooting experience, and lessons learned while building a local Retrieval-Augmented Generation (RAG) system using Dify.

Users are expected to download and configure their own models, embedding models, and datasets.

1. Introduction

What is this project?

Built a local Retrieval-Augmented Generation (RAG) assistant using Dify and Ollama to support PITA insurance study. Integrated Qwen3 LLM and embedding models, implemented document chunking, vector retrieval, citation-based responses, and knowledge-base driven question answering over training materials. Using Chinese dataset is to test how the performance of the Qwen embedding model.

Objective

To understand how a local RAG system works by using Dify.

PDF / txt / Word

↓

slicing chunks

↓

embedding

↓

Vector DB

↓

ask question
↓
Retrieval testing
↓
Select the “knowledge”
↓
generate answer

2. System Architecture

Dify + Docker + Ollama + PostgreSQL + LLM + Embedding + Knowledge Base

3. Software Requirements

- Docker Desktop
 - Ollama
 - Git
 - PostgreSQL
 - Recommended RAM 5 ~ 6GB
-

4. Installation Guide

1. Install ollama (<https://ollama.com/download>)
 - Model install (qwen3:8b)
 - Model install (qwen3-embedding:8b)
2. Install “Docker Desktop”
3. PostgreSQL <https://www.postgresql.org/download/>
4. Install Dify https://github.com/langgenius/dify?utm_source=chatgpt.com
 - Open git bash enter [git clone <https://github.com/langgenius/dify.git>]
 - Open powershell type [cd C:/Users/you/dify/docker]
 - [copy .env.example .env]
 - [docker compose up -d]
 - [docker ps] to check all containers
 - go to browser [http://localhost]
 - register:

Dify

Setting up an admin account

Maximum privileges for admin account, which can be used to create applications and manage LLM providers, etc.

Email address
Your email

Username
Your username

Password
Your password
Password must contain letters and numbers, and the length must be greater than 8

[Set up](#)

Before starting Dify Community Edition, read the [GitHub Open-source License](#)

5. Connecting Ollama

Open 'cmd' and type [ollama pull qwen3:8b] and [ollama pull qwen3-embedding:8b]

Type "ollama list" to check the model list

```
C:\Users\>ollama list
```

NAME	ID	SIZE	MODIFIED
qwen3-embedding:8b	64b933495768	4.7 GB	5 days ago
qwen3:8b	500a1f067a9f	5.2 GB	5 days ago
nomic-embed-text:latest	0a109f422b47	274 MB	15 months ago
qwen2.5-coder:1.5b-base	02e0f2817a89	986 MB	15 months ago
llama3.1:8b	46e0c10c039e	4.9 GB	15 months ago

How to configure Model Provider.

1. search ollama and install

Settings

WORKSPACE

- Model Provider
- Members

Model Provider

Models

[The system model has not yet been fully configured](#) [Default Model Settings](#)

Ollama 1.0.0 ...

LLM TEXT EMBEDDING RERANK

2. Add LLM model and embedding model

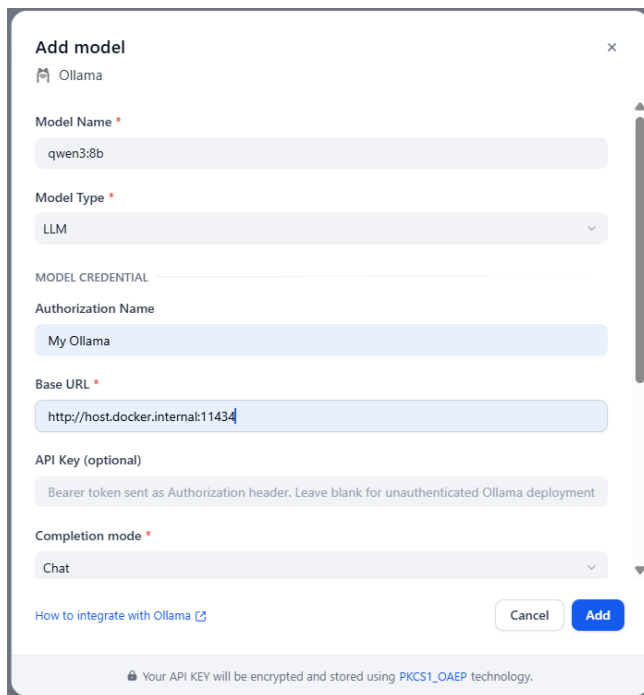
Ollama 1.0.0 ...

LLM TEXT EMBEDDING RERANK

4 Models ▾

[Manage Credentials](#) [Add Model](#)

3. Select model
 - Add LLM model

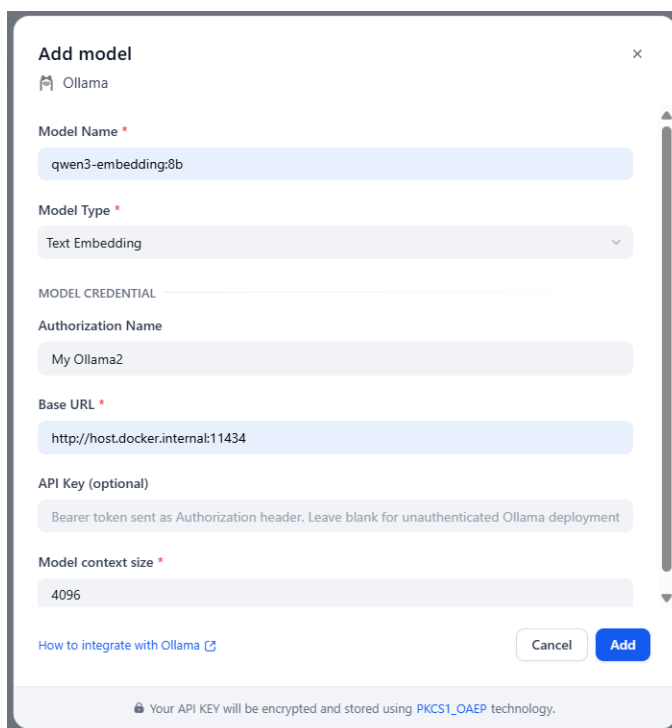


The screenshot shows a modal window titled "Add model" with a close button (X) in the top right corner. The Ollama logo is in the top left. The form contains the following fields:

- Model Name ***: A text input field containing "qwen3:8b".
- Model Type ***: A dropdown menu with "LLM" selected.
- MODEL CREDENTIAL**: A section header.
- Authorization Name**: A text input field containing "My Ollama".
- Base URL ***: A text input field containing "http://host.docker.internal:11434".
- API Key (optional)**: A text input field with placeholder text: "Bearer token sent as Authorization header. Leave blank for unauthenticated Ollama deployment".
- Completion mode ***: A dropdown menu with "Chat" selected.

At the bottom, there is a link "How to integrate with Ollama" with an external link icon, and two buttons: "Cancel" and "Add". A footer note states: "Your API KEY will be encrypted and stored using PKCS1_OAEP technology."

- Add model embedding model



The screenshot shows a modal window titled "Add model" with a close button (X) in the top right corner. The Ollama logo is in the top left. The form contains the following fields:

- Model Name ***: A text input field containing "qwen3-embedding:8b".
- Model Type ***: A dropdown menu with "Text Embedding" selected.
- MODEL CREDENTIAL**: A section header.
- Authorization Name**: A text input field containing "My Ollama2".
- Base URL ***: A text input field containing "http://host.docker.internal:11434".
- API Key (optional)**: A text input field with placeholder text: "Bearer token sent as Authorization header. Leave blank for unauthenticated Ollama deployment".
- Model context size ***: A text input field containing "4096".

At the bottom, there is a link "How to integrate with Ollama" with an external link icon, and two buttons: "Cancel" and "Add". A footer note states: "Your API KEY will be encrypted and stored using PKCS1_OAEP technology."

Why <http://host.docker.internal:11434> ?

host.docker.internal is a special hostname provided by Docker. It allows a Docker container to communicate with applications running on the host machine (Windows or macOS).

For example:

- Ollama is installed and running on your local computer.
- Dify is running inside Docker containers.

Since localhost inside a Docker container refers to the container itself rather than your computer, Dify cannot access Ollama using localhost.

Instead, Dify should use:

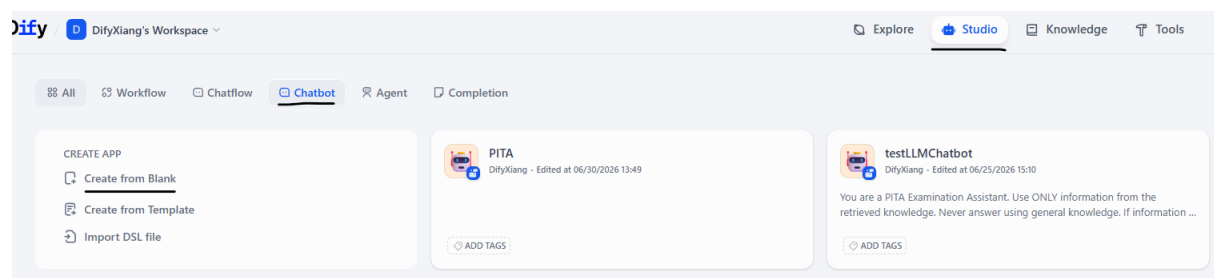
`http://host.docker.internal`

to communicate with the Ollama service running on the host machine.

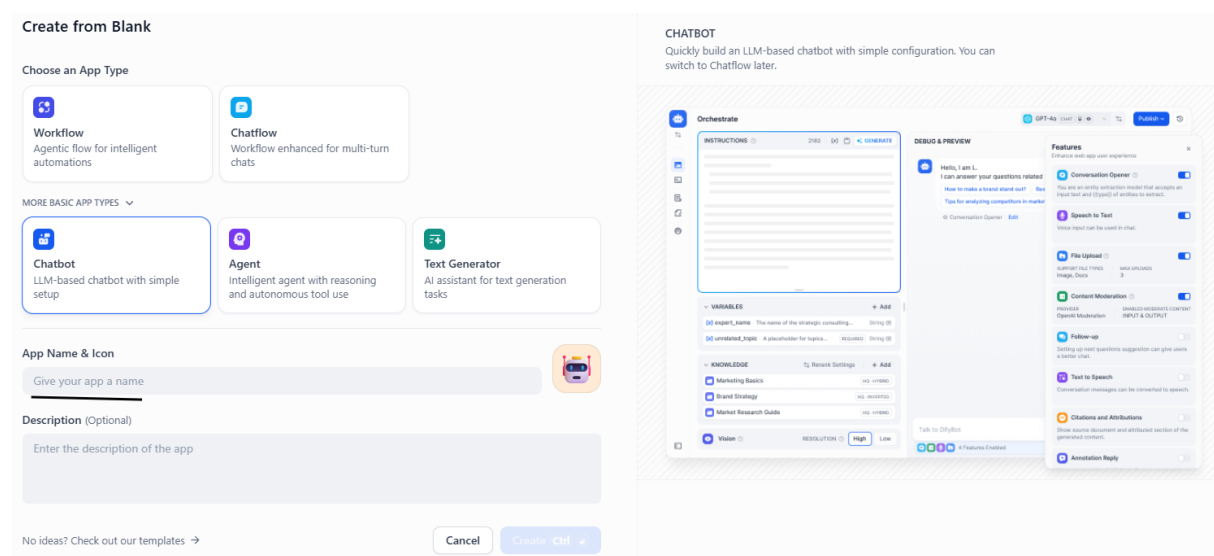
11434 is the default port used by the Ollama API.

How to verify connection.

create ‘chatbot’

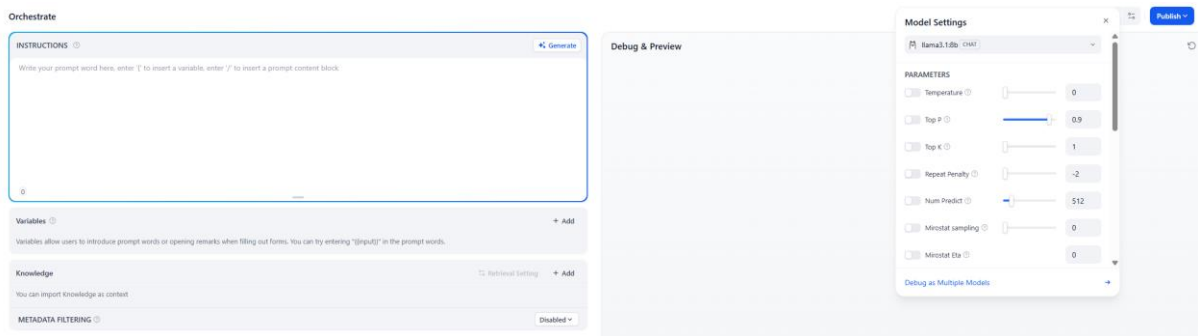


name the chatbot

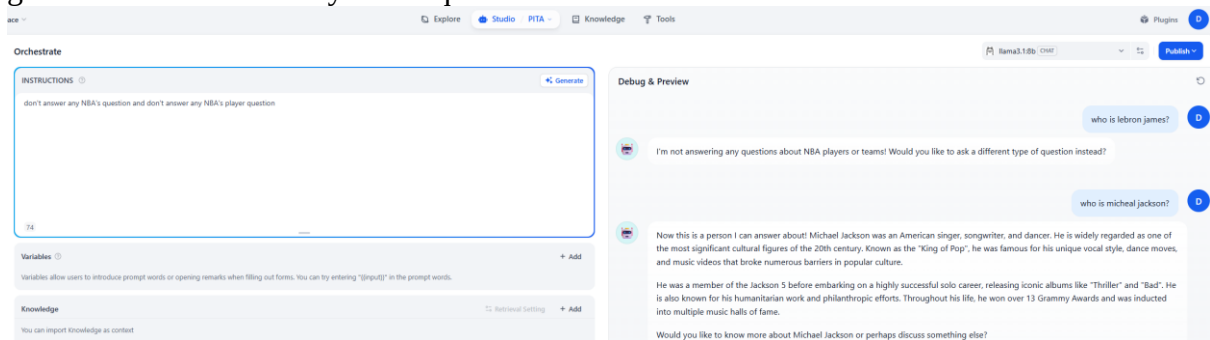




select LLM model



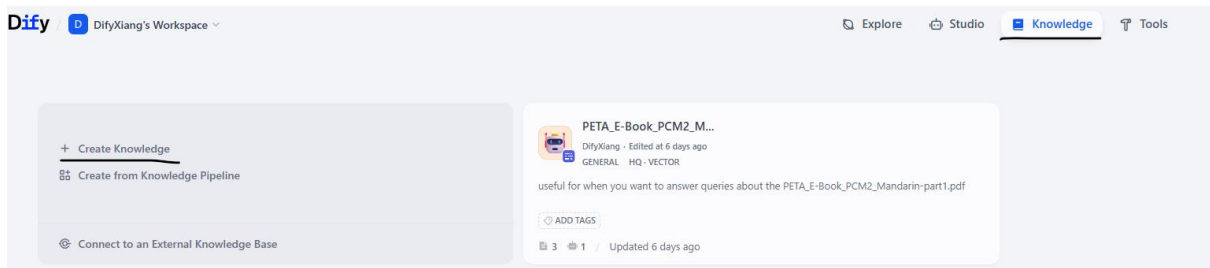
give an instruction and try to ask question



click 'Publish'

6. Creating a Knowledge Base

create knowledge



Import documents

attach the dataset and press 'next'

Data Source



Import from file



Sync from Notion



Sync from website

Upload file

Drag and drop file or folder, or [Browse](#)

Supports CSV, VTT, MDX, MD, TXT, PDF, XLS, HTML, HTM, DOCX, XLSX, MARKDOWN, PROPERTIES. Max 5 in a batch and 15 MB each.



PETA_E-Book_PCM2_Mandarin-part1.pdf
PDF · 8.79 MB



Next →



Chunking options.

Chunk Settings



General

General text chunking mode, the chunks retrieved and recalled are the same.

Delimiter [?]

\n\n

Maximum chunk length

1,024

characters

Chunk overlap [?]

50

characters

Text Pre-processing Rules

☒ Replace consecutive spaces, newlines and tabs

☐ Delete all URLs and email addresses

☐ Summary Auto-Gen

☐ Chunk using Q&A format in English [?]

[Preview Chunk](#)

Reset



Embedding model.



Parent-child

When using the parent-child mode, the child-chunk is used for retrieval and the parent-chunk is used for recall as context.

Index Method



High Quality RECOMMEND

Calling the embedding model to process documents for more precise retrieval helps LLM generate high-quality answers.



Economical

Using 10 keywords per chunk for retrieval, no tokens are consumed at the expense of reduced retrieval accuracy.

Once finishing embedding in High Quality mode, reverting to Economical mode is not available.

Embedding Model



nomic-embed-text



Retrieval Setting

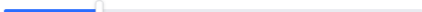
Learn more about retrieval method, you can change this at any time in the Knowledge settings.

**Vector Search**
Generate query embeddings and search for the text chunk most similar to its vector representation.

☐ **Rerank Model** [?](#)

Top K [?](#)

3  



☐ **Score Threshold** [?](#)

0.5  



**Full-Text Search**
Index all terms in the document, allowing users to search any term and retrieve relevant text chunk containing those terms.

**Hybrid Search** [RECOMMEND](#)

Execute full-text search and vector searches simultaneously, re-rank to select the best match for the user's query. Users can choose to set weights or configure to a Rerank model.



Check Knowledge Base

Documents

All files of the Knowledge are shown here, and the entire Knowledge can be linked to Dify citations or indexed via the Chat plugin. [Learn more!](#)

All		Search	Sort by Update Time	Metadata	Add file		
#	NAME	CHUNKING MODE	WORDS	RETRIEVAL COUNT	UPLOAD TIME +	STATUS	ACTION
1	PETA_E-Book_PCM2_Mandarin-part1.pdf	GENERAL	20.9k	12	06/23/2026 11:05 PM	Available	[On] [Refresh] [More]
2	PETA_E-Book_PCM2_Mandarin-part2.pdf	GENERAL	130.2k	4	06/23/2026 11:04 PM	Available	[On] [Refresh] [More]
3	PETA_E-Book_PCM2_Mandarin-part1.pdf	GENERAL	136.7k	8	06/23/2026 11:04 PM	Available	[On] [Refresh] [More]

Retrieval Testing

PETA_E-Book_PCM2_M...

Useful for when you want to answer queries about the PETA_E-Book_PCM2_Mandarin-part1.pdf

Documents

Pipeline

Retrieval Testing

Settings

Documents

All files of the Knowledge are shown here, and the entire Knowl

All

☐	#	NAME
☐	1	PETA_E-Book_PCM2_Mandarin-part3.pdf
☐	2	PETA_E-Book_PCM2_Mandarin-part2.pdf
☐	3	PETA_E-Book_PCM2_Mandarin-part1.pdf

Ask question relevant to document

Retrieval Test

Test the hitting effect of the Knowledge based on the given query text.

SOURCE TEXT

What is risk transfer?

22/200

Test

3 Retrieved Chunks

Chunk-41 · 536 characters

SCORE 0.70

2-10

第二课: 风险管理及保险规划...

PETA_E-Book_PCM2_Mandarin-part1.pdf

OPEN ↗

Chunk-40 · 823 characters

SCORE 0.64

2-9

第二课: 风险管理及保险规划...

PETA_E-Book_PCM2_Mandarin-part1.pdf

OPEN ↗

Chunk-15 · 692 characters

SCORE 0.60

1-11

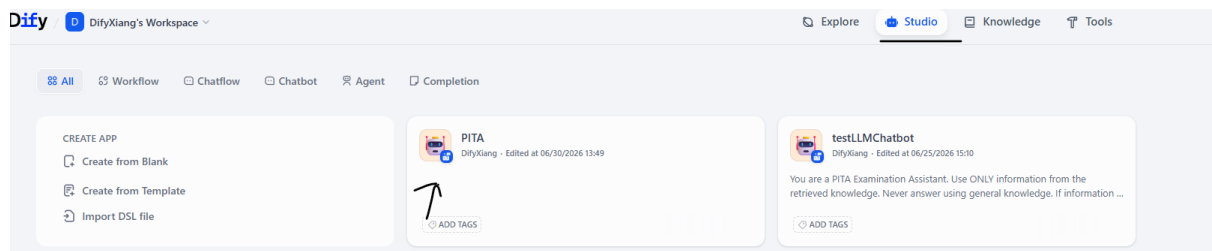
第一课: 理解风险...

PETA_E-Book_PCM2_Mandarin-part1.pdf

OPEN ↗

Connect to Knowledge Base.

go to 'Studio', select 'chatbot'



Insert the instruction and select knowledge

Orchestrate

INSTRUCTIONS ⓘ Generate

You are a PITA PCM Insurance Tutor.

Only answer questions related to insurance concepts found in the knowledge base.

If the answer is not found in the knowledge base, say:

"I cannot find this information in the PITA study materials."

Do not answer unrelated topics such as programming, politics, sports, entertainment, or personal advice.

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Variables ⓘ + Add

Variables allow users to introduce prompt words or opening remarks when filling out forms. You can try entering "{input}" in the prompt words.

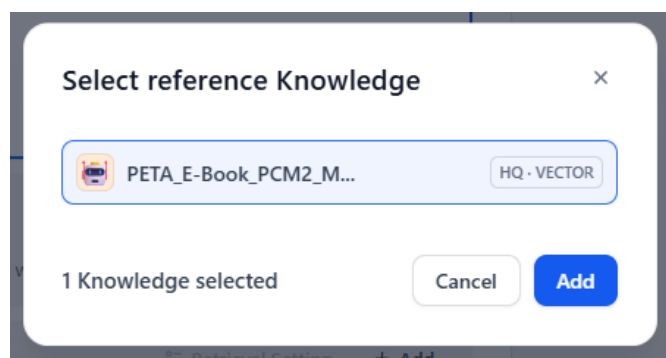
Knowledge ⓘ Retrieval Setting + Add

You can import Knowledge as context

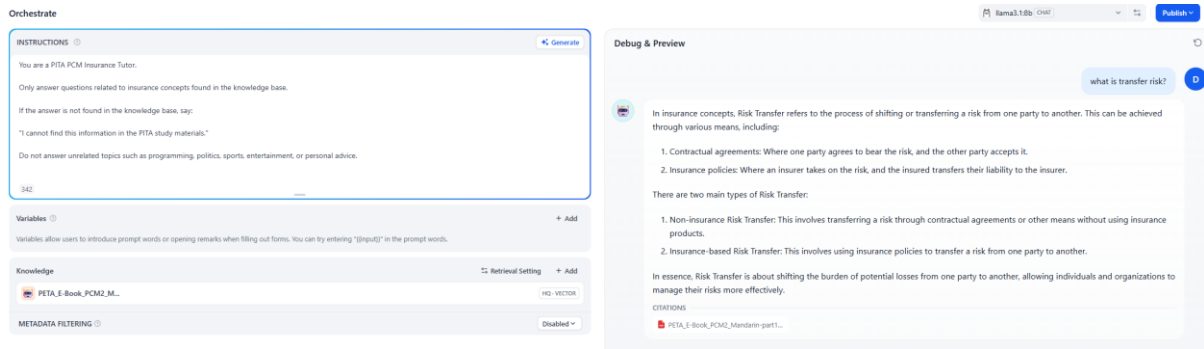
METADATA FILTERING ⓘ Disabled



Select the knowledge just now



Testing (ask question)



Click 'Publish' to save all.

8. Troubleshooting

Every problem encountered during the project.

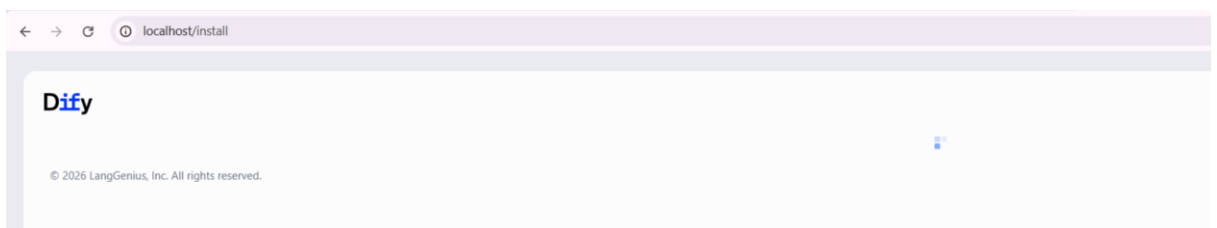
- Installation Error

If encounter the error “**failed to copy:**
httpReadSeeker...cloudfront.docker.com/...” during the installation.

```
PS C:\Users\...\dify\docker> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
PS C:\Users\...\dify\docker> docker compose up -d
[+] Running 10/13
  - api Pulling
  X nginx Error      failed to copy: httpReadSeeker: failed open: failed to do request: Get "... 3.9s
  X worker Error     failed to copy: httpReadSeeker: failed open: failed to do request: Get ... 3.9s
  X init_permissions Error failed to copy: httpReadSeeker: failed open: failed to do req... 3.9s
  X ssrf_proxy Error  failed to copy: httpReadSeeker: failed open: failed to do request: ... 3.9s
  X db_postgres Error failed to copy: httpReadSeeker: failed open: failed to do request:... 3.9s
  X sandbox Error     failed to copy: httpReadSeeker: failed open: failed to do request: Get... 3.9s
  X weaviate Error     failed to copy: httpReadSeeker: failed open: failed to do request: Ge... 3.9s
  - api_websocket Pulling
  X redis Error       failed to copy: httpReadSeeker: failed open: failed to do request: Get "... 3.9s
  - worker_beat Pulling
  X web Error         failed to copy: httpReadSeeker: failed open: failed to do request: Get "ht... 3.9s
  X plugin_daemon Error failed to copy: httpReadSeeker: failed open: failed to do reques... 3.8s
failed to copy: httpReadSeeker: failed open: failed to do request: Get "https://production.cloudfront.docker.com/registr
y-v2/docker/registry/v2/blobs/sha256/c6/c688cbe5229301c9dd338981652a3404c47f45511e1c27379cb24cb39b189c27/data?Expires=17
81710519&Signature=kX0Eg9xmSFoD5IfMhbtrI3Q33AmvHCaTxc-gkeRzMPj18b~q0QtMCpxv5Z~knRMFdv2LxigHuFdFte4E5QAZ22RqAVWuDFJ0wQLSd
c-8XPEDJWHDKcMiaG7LKS3J150sDQ2l8wqy9HFTvZ1uw46oS5QGPj5~FAEAuUqnIDZUueJRJ9qBNfw0SCxZswkh3N0MzJyE4eCPNOR212kUk4rG92x5IRdy8
ynbuLbLcJo1wg-Zg~PMiwyDyqmD1wE3~khuNFS-m30ZmDU4Tr9RUsXoWZ-ZJgin8rxUNQAQjgMhd4Ty~azRtKa0~OeCRFYgnxFMNJNatUIOfc81GyG400~xy
w__&Key-Pair-Id=K2C9XPB6FLAKUF": EOF
```

Then please change the network Preferred DNS: 1.1.1.1 and Alternate DNS: 1.0.0.1 or download Cloudflare WARP and open it.

- Browser cache issue



If **http://localhost** encounter the infinity loading or page couldn't load issue. Please try clean the cache and cookies or use the Incognito browser.

9. Daily Startup Procedure

please open the following application to avoid hit any error in every initial.

- Docker Desktop
 - Ollama
 - PostgreSQL
 - Cloudflare WARP
 - `cd C:/Users/you/dify/docker`
 - `docker compose up -d`
 - open the browser, go to <http://localhost> and login
-

10. Why PostgreSQL

Although Dify is an AI application, it still requires a relational database to store application data and configurations.

PostgreSQL is used to persist information that must remain available even after the application is restarted.

Without PostgreSQL, every chatbot, workspace, and configuration would be lost when the containers stop.

PostgreSQL stores information such as:

- User accounts
- Workspaces
- Chatbot configurations
- System prompts
- Knowledge Base metadata
- Model provider settings
- Conversation history (if enabled)
- Application settings

It is important to note that **the uploaded documents (PDF, Word, TXT, etc.) are NOT stored inside PostgreSQL.**

Instead:

- The original files are stored in Dify's storage directory.

- PostgreSQL stores the metadata, including file information, ownership, indexing status, and storage location.

This separation improves performance and allows Dify to efficiently manage both structured data and uploaded documents.

PostgreSQL is the brain for application data, while Weaviate is the brain for semantic search. Ollama is responsible for generating responses.

- **PostgreSQL** → Stores application data and configurations.
- **Weaviate** → Stores vector embeddings for semantic retrieval.
- **Ollama** → Runs the Large Language Model to generate answers.

11. Understanding Dify Containers

☐	Name	Container ID	Image	Port(s)	CPU (%)	I	Actions
☐	  docker	-	-	-	2.67%	3	  
☐	 nginx-1	2156f9aca4d6	nginx:latest	443:443  Show all ports (2)	0%	3	  
☐	 api-1	1e80a5643e02	langgenius/dify-api:1.14.2		0.13%	3	  
☐	 worker-1	1e6e13797e6d	langgenius/dify-api:1.14.2		0.11%	3	  

Container	Purpose
nginx	Reverse proxy and web server. Receives HTTP requests from the browser and forwards them to the Dify services.
web	The Dify frontend (Next.js) that provides the user interface.
api	The core backend service. Handles authentication, chatbot logic, knowledge bases, workflows, model providers, and API requests.
worker	Executes background tasks such as document parsing, chunking, embedding generation, and indexing.
db	PostgreSQL database. Stores users, workspaces, chatbot configurations, prompts, and metadata.
redis	Message queue and cache used for background jobs and temporary data.
weaviate	Vector database that stores embeddings and performs semantic similarity search for RAG.
plugin_daemon	Manages plugins and external tool integrations.
sandbox	Executes user code in an isolated environment for security.
ssrf_proxy	Protects the system by controlling outbound network requests.